

$$f : \mathbb{R}^4 \rightarrow \mathbb{R}$$

$$\max f(x) = 3x_1 + 2x_2 + 5x_3 + 4x_4$$

$$x_1 + x_2 + x_3 + x_4 \leq 10$$

$$2x_1 + x_3 \leq 8$$

$$x_2 + 2x_4 \leq 7$$

$$x_3 + x_4 \leq 6$$

$$3x_1 + x_2 \leq 9$$

$$x_1, x_2, x_3, x_4 \geq 0$$

Forma standard

$$\min f(x) = -3x_1 - 2x_2 - 5x_3 - 4x_4$$

$$x_1 + x_2 + x_3 + x_4 + s_1 = 10$$

$$2x_1 + x_3 + s_2 = 8$$

$$x_2 + 2x_4 + s_3 = 7$$

$$x_3 + x_4 + s_4 = 6$$

$$3x_1 + x_2 + s_5 = 9$$

$$x_1, x_2, x_3, x_4, s_1, s_2, s_3, s_4, s_5 \geq 0$$

Metodo del simplesso:

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	1	1	1	1	1	0	0	0	0	10
s_2	2	0	1	0	0	1	0	0	0	8
s_3	0	1	0	2	0	0	1	0	0	7
s_4	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	-5	-4	0	0	0	0	0	0

Entra x_3 ed esce s_4 :

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	1	1	1	1	1	0	0	0	0	10
s_2	2	0	1	0	0	1	0	0	0	8
s_3	0	1	0	2	0	0	1	0	0	7
s_4	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	-5	-4	0	0	0	0	0	0

$R1 \leftarrow R1 - R4$

$$\begin{array}{cccccccc|c} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 10 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 6 \\ \hline 1 & 1 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 4 \end{array}$$

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	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	1	1	0	0	1	0	0	-1	0	4
s_2	2	0	1	0	0	1	0	0	0	8
s_3	0	1	0	2	0	0	1	0	0	7
s_4	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	-5	-4	0	0	0	0	0	0

$R2 \leftarrow R2 - R4$

$$\begin{array}{cccccccc|c} 2 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 8 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 6 \\ \hline 2 & 0 & 0 & -1 & 0 & 1 & 0 & -1 & 0 & 2 \end{array}$$

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	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	1	1	0	0	1	0	0	-1	0	4
s_2	2	0	0	-1	0	1	0	-1	0	2
s_3	0	1	0	2	0	0	1	0	0	7
s_4	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	-5	-4	0	0	0	0	0	0

$$R6 \leftarrow R6 + R4 \cdot 5$$

$$\begin{array}{cccccccc|c} -3 & -2 & -5 & -4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 5 & 0 & 0 & 0 & 5 & 0 & 30 \\ \hline -3 & -2 & 0 & -1 & 0 & 0 & 0 & 5 & 0 & 30 \end{array}$$

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	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	1	1	0	0	1	0	0	-1	0	4
s_2	2	0	0	-1	0	1	0	-1	0	2
s_3	0	1	0	2	0	0	1	0	0	7
x_3	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	0	-1	0	0	0	5	0	30

Entra x_1 ed esce s_2 :

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	1	1	0	0	1	0	0	-1	0	4
s_2	2	0	0	-1	0	1	0	-1	0	2
s_3	0	1	0	2	0	0	1	0	0	7
x_3	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	0	-1	0	0	0	5	0	30

$$R2 \leftarrow R2 \cdot \frac{1}{2}$$

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	1	1	0	0	1	0	0	-1	0	4
s_2	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	1	0	2	0	0	1	0	0	7
x_3	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	0	-1	0	0	0	5	0	30

$$R1 \leftarrow R1 - R2$$

$$\begin{array}{cccccc|c} 1 & 1 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 4 \\ 1 & 0 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 1 \\ \hline 0 & 1 & 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 & -\frac{1}{2} & 0 & 3 \end{array}$$

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	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	0	1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	3
s_2	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	1	0	2	0	0	1	0	0	7
x_3	0	0	1	1	0	0	0	1	0	6
s_5	3	1	0	0	0	0	0	0	1	9
	-3	-2	0	-1	0	0	0	5	0	30

$$R5 \leftarrow R5 - R2 \cdot 3$$

$$\begin{array}{cccccccc|c}
3 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 9 \\
3 & 0 & 0 & -\frac{3}{2} & 0 & \frac{3}{2} & 0 & -\frac{3}{2} & 3 \\
\hline
0 & 1 & 0 & \frac{3}{2} & 0 & -\frac{3}{2} & 0 & \frac{3}{2} & 6
\end{array}$$

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	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	0	1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	3
s_2	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	1	0	2	0	0	1	0	0	7
x_3	0	0	1	1	0	0	0	1	0	6
s_5	0	1	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	0	$\frac{3}{2}$	1	6
	-3	-2	0	-1	0	0	0	5	0	30

$$R6 \leftarrow R6 + R2 \cdot 3$$

$$\begin{array}{cccccccc|c}
-3 & -2 & 0 & -1 & 0 & 0 & 0 & 5 & 0 & 30 \\
3 & 0 & 0 & -\frac{3}{2} & 0 & \frac{3}{2} & 0 & -\frac{3}{2} & 0 & 3 \\
\hline
0 & -2 & 0 & \frac{1}{2} & 0 & \frac{3}{2} & 0 & \frac{7}{2} & 0 & 33
\end{array}$$

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	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	0	1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	3
x_1	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	1	0	2	0	0	1	0	0	7
x_3	0	0	1	1	0	0	0	1	0	6
s_5	0	1	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	0	$\frac{3}{2}$	1	6
	0	-2	0	$\frac{1}{2}$	0	$\frac{3}{2}$	0	$\frac{7}{2}$	0	33

Entra x_2 ed esce s_1 :

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	0	1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	3
x_1	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	1	0	2	0	0	1	0	0	7
x_3	0	0	1	1	0	0	0	1	0	6
s_5	0	1	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	0	$\frac{3}{2}$	1	6
	0	-2	0	$\frac{1}{2}$	0	$\frac{3}{2}$	0	$\frac{7}{2}$	0	33

$$R3 \leftarrow R3 - R1$$

$$\begin{array}{cccccc|c}
0 & 1 & 0 & 2 & 0 & 0 & 1 & 0 & 0 & 7 \\
0 & 1 & 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 & -\frac{1}{2} & 0 & 3 \\
\hline
0 & 0 & 0 & \frac{3}{2} & -1 & \frac{1}{2} & 1 & \frac{1}{2} & 0 & 4
\end{array}$$

↓

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	0	1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	3
x_1	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	0	0	$\frac{3}{2}$	-1	$\frac{1}{2}$	1	$\frac{1}{2}$	0	4
x_3	0	0	1	1	0	0	0	1	0	6
s_5	0	1	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	0	$\frac{3}{2}$	1	6
	0	-2	0	$\frac{1}{2}$	0	$\frac{3}{2}$	0	$\frac{7}{2}$	0	33

$$R5 \leftarrow R5 - R1$$

$$\begin{array}{cccccc|c}
0 & 1 & 0 & \frac{3}{2} & 0 & -\frac{3}{2} & 0 & \frac{3}{2} & 1 & 6 \\
0 & 1 & 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 & -\frac{1}{2} & 0 & 3 \\
\hline
0 & 0 & 0 & 1 & -1 & -1 & 0 & 2 & 1 & 3
\end{array}$$

↓

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
s_1	0	1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	3
x_1	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	0	0	$\frac{3}{2}$	-1	$\frac{1}{2}$	1	$\frac{1}{2}$	0	4
x_3	0	0	1	1	0	0	0	1	0	6
s_5	0	0	0	1	-1	-1	0	2	1	3
	0	-2	0	$\frac{1}{2}$	0	$\frac{3}{2}$	0	$\frac{7}{2}$	0	33

$$R6 \leftarrow R6 + R1 \cdot 2$$

$$\begin{array}{cccccc|c} 0 & -2 & 0 & \frac{1}{2} & 0 & \frac{3}{2} & 0 & \frac{7}{2} & 0 & 33 \\ 0 & 2 & 0 & 1 & 2 & -1 & 0 & -1 & 0 & 6 \\ \hline 0 & 0 & 0 & \frac{3}{2} & 2 & \frac{1}{2} & 0 & \frac{5}{2} & 0 & 39 \end{array}$$

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	x_1	x_2	x_3	x_4	s_1	s_2	s_3	s_4	s_5	b
x_2	0	1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	3
x_1	1	0	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	1
s_3	0	0	0	$\frac{3}{2}$	-1	$\frac{1}{2}$	1	$\frac{1}{2}$	0	4
x_3	0	0	1	1	0	0	0	1	0	6
s_5	0	0	0	1	-1	-1	0	2	1	3
	0	0	0	$\frac{3}{2}$	2	$\frac{1}{2}$	0	$\frac{5}{2}$	0	39

$$x_1 = 1$$

$$x_2 = 3$$

$$x_3 = 6$$

$$\min f(x) = -3x_1 - 2x_2 - 5x_3 - 4x_4 \rightarrow f(x) = -3(1) - 2(3) - 5(6) - 4(0) = -3 - 6 - 30 = -39$$

$$\max f(x) = 3x_1 + 2x_2 + 5x_3 + 4x_4 \rightarrow \boxed{39}$$